

BABU BANARASI DAS UNIVERSITY, LUCKNOW

Department of Computer Applications • Department of Data Science & AI

MAJOR ACADEMIC PROJECT REPORT

CAMPUS PLACEMENT & RECRUITMENT TRACKER

A High-Performance Document-Oriented Management & Analytics System

Powered by MongoDB Community Server 8.3 & Java Synchronous Driver

Submitted in partial fulfillment of the requirements for the degree of:

BACHELOR OF COMPUTER APPLICATIONS

(Data Science & Artificial Intelligence)

Submitted By:

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University Roll No: 26 / 12502

Course / Branch: BCA (Data Science & AI)

Academic Session: 2025 – 2026

Lucknow, Uttar Pradesh, India

CERTIFICATE OF DECLARATION

I hereby declare that this academic project report entitled "**Campus Placement & Recruitment Tracker**" submitted to the Department of Computer Applications, Babu Banarasi Das University, Lucknow, is an authentic record of my technical work carried out under proper academic supervision. All database implementations, Java DAOs, aggregation pipelines, and graphical dashboards have been validated against a live MongoDB Community Server 8.3 deployment.

Date: September 24, 2026

Place: Lucknow

Shaswat Jaiswal

Roll No: 26 / 12502

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ACKNOWLEDGEMENTS

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1. EXECUTIVE SUMMARY & OBJECTIVES

The **Campus Placement & Recruitment Tracker** is a document-oriented software system designed to optimize higher-education placement cell administration. Engineered with **MongoDB Community Server 8.3** and the **Java Synchronous Driver (v5.1.1)**, the application addresses the stiffness of legacy relational schemas by supporting nested multikey attributes (such as candidate technical skills) and execution of multi-stage aggregation pipelines directly within the database engine.

Core Technical Objectives:

- Scalable Schema Modeling:** Normalized collections (students, companies, applications) with B-Tree indexes.
- Production Ingestion:** Implement insertOne for manual registration and bulk insertMany loading 72 production records (exceeding the >=65 threshold).
- Dynamic Filtering:** Support corporate cutoff screening (\$gte, \$gt, \$lte), range queries, array matches (\$in, \$all), and complex logic (\$and, \$or).
- Institutional Analytics:** Real-time KPI aggregation pipelines computing departmental placement rates, average/max packages, and top scorers.
- Performance Validation:** Benchmarking query execution plans via .explain("executionStats").

2. MONGODB OPERATORS REFERENCE MATRIX

Category	Operator	BSON Syntax	System Application
Comparison	\$eq	<code>{ roll_number: '26 / 12502' }</code>	Exact Roll Number candidate lookup
Comparison	\$gte	<code>{ cgpa: { \$gte: 8.0 } }</code>	Corporate cutoff screening for eligibility
Comparison	\$gt	<code>{ cgpa: { \$gt: 9.0 } }</code>	Elite tier-1 candidate filtering
Comparison	\$lte / \$lt	<code>{ cgpa: { \$lte: 7.0 } }</code>	Remedial and upper-bound filtering
Comparison	\$ne	<code>{ is_placed: { \$ne: true } }</code>	Retrieving unplaced student pool
Logical	\$and	<code>{ \$and: [{ cgpa: { \$gte: 8 } }, ...] }</code>	Enforcing composite eligibility criteria
Logical	\$or	<code>{ \$or: [{ dept: 'BCA DS & AI' }, ...] }</code>	Multi-department recruitment eligibility
Array	\$in	<code>{ skills: { \$in: ['Python', 'SQL'] } }</code>	Matching candidates with any target skill
Array	\$all	<code>{ skills: { \$all: ['Java', 'MongoDB'] } }</code>	Strict multi-skill prerequisite check
Evaluation	\$regex	<code>{ name: { \$regex: 'Shaswat', \$options: 'i' } }</code>	Case-insensitive name search

3. REAL-TIME PLACEMENT ANALYTICS & KPI METRICS

Institutional metrics were evaluated on the live 72-record database through multi-stage aggregation pipelines:

Metric Description	Aggregated Value	Metric Description	Aggregated Value
Total Candidates Seeded	72 (Threshold: >=65)	Placed Candidates	46 Students
Unplaced Candidates	26 Students	Placement Success Rate	63.89%
Institutional Avg CGPA	8.02 / 10.0	Highest Package Offered	22.0 LPA (Microsoft)
Average Package (Placed)	11.54 LPA	Star Profile	Shaswat Jaiswal (18.5 LPA)

Department-Wise Placement Statistics (\$group, \$cond, \$project):

Department	Total	Placed	Unplaced	Placement %	Avg Package	Max Package
BCA DS & AI	19	12	7	63.16%	13.56 LPA	20.0 LPA
B.Tech CSE	20	14	6	70.00%	12.94 LPA	22.0 LPA
MCA	11	7	4	63.64%	9.29 LPA	14.0 LPA
B.Tech IT	12	8	4	66.67%	11.45 LPA	20.0 LPA
B.Tech ECE	10	5	5	50.00%	10.20 LPA	22.0 LPA

Top Academic Performers:

- Vikas Dubey — B.Tech CSE | CGPA: 9.60 | Placed: Microsoft (22.0 LPA)
- Shaswat Jaiswal — BCA DS & AI | CGPA: 9.48 | Placed: Google Cloud (18.5 LPA)
- Hemant Kulkarni — MCA | CGPA: 9.30 | Placed: Oracle (12.0 LPA)

4. QUERY PROFILING & CONCLUSION

MongoDB Explain Plan Output (.explain('executionStats')):

Execution Time Millis : 5 ms Docs Returned (n) : 22 Total Keys Examined : 22 Total Docs Examined : 22 Execution Plan Stage : FETCH <- IXSCAN on index 'cgpa_-1' Assessment : Optimal B-Tree index scan; zero unnecessary documents examined.

Academic Project Conclusion:

The Campus Placement & Recruitment Tracker successfully satisfies all technical and structural guidelines prescribed for the BCA (Data Science & AI) degree at Babu Banarasi Das University. By uniting MongoDB 8.3's flexible document paradigm with Java's robust synchronous driver, the system provides high performance, real-time intelligence, and an intuitive dual-interface platform for institutional placement administration.